

First Problem Sheet

on

General Topology

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Topic: Metric Spaces

"All of mathematics is some kind of metric space."
-Israel Gelfand

Problem 1

Let $x \in \mathbb{R}^n$ and let $s = \sup\{\pi^j(x)\}_{j=1}^n$. Show that

$$s \leq |x| \leq s\sqrt{n}.$$

Problem 2

Suppose M is a metric space.

- i) Show that an open ball in M is an open subset, and a closed ball in M is a closed subset.
- ii) Show that a subset of M is open if and only if it is the union of some collection of open balls.

Problem 3

Let M and N be metric spaces and let $f : M \rightarrow N$ be a map. Show that f is continuous if and only if it takes convergent sequences to convergent sequences and limits to limits, that is, if and only if $x_n \rightarrow x$ in M implies $f(x_n) \rightarrow f(x)$ in N .

Problem 4

Suppose A is a closed subset of a metric space M and (x_n) is a sequence of points in A that converges to a point $x \in M$. Show that $x \in A$.

Problem 5

Prove that every convergent sequence in a metric space is Cauchy, and every Cauchy sequence is bounded.

Problem 6

Prove the open subset criterion for continuity.